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SIREN

Sepsis Infection Risk Early Notification

[TEAM NAME]

Track: Healthcare Digital Transformation

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Team members

[Student name]

Clinical lead · Year [x] medical student

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Data and prototype lead · Year [x] medical student

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Design and user research lead · Year [x] medical student

Mentored by a clinical pharmacist and PhD researcher in AI for sepsis (advisory role only; the idea, prototype and presentation are the team's own).

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Problem Statement

Sepsis is the body's life-threatening response to infection. It kills about 11 million people a year worldwide and is involved in one in every two to three hospital deaths. Every hour of delay in treatment raises the odds of death by about 4 %.

11 M

deaths a year worldwide (Rudd, Lancet 2020)

1 in 2–3

hospital deaths involve sepsis (Liu, JAMA 2014)

+4 % / h

odds of death per hour of treatment delay (Seymour, NEJM 2017)

40 %

ICU mortality among septic patients in one Saudi hospital (Buraidah, 2012)

Stakeholders: ICU and ward patients, nurses, physicians, hospital quality teams, the Ministry of Health. The problem is not detection in theory; it is detection early enough, in every hospital, without drowning clinicians in alarms.

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Why today's sepsis alerts fail

Missed cases, constant alarms

The most widely deployed hospital sepsis model missed 67 % of sepsis cases and alerted on 18 % of all admitted patients.

Wong et al., JAMA Intern Med 2021

Models that do not travel

The winning model of the PhysioNet 2019 challenge scored 0.43 on its own hospitals and −0.12, worse than silence, on a hospital it had never seen.

Reyna et al., Crit Care Med 2020

Tested on themselves

81 % of published clinical AI models were never tested on data from another hospital; most of the rest lost performance when they were.

Yu et al., Radiol Artif Intell 2022

Three root causes we read in the literature

1. Every hospital stores the same vital sign under a different name, unit and clock, so a model trained in one hospital reads nonsense in the next.
2. Models are often trained with information from after the moment of prediction (a test ordered because the doctor already suspected sepsis), so they look brilliant on paper and fail at the bedside.
3. They are judged by a statistic (AUROC) that counts neither the patients missed nor the alarms clinicians have to answer.

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Innovation Overview

SIREN is an early sepsis warning for the whole hospital that speaks every ward's data language, uses only what the clinician could have known at that hour, and is judged by patients helped and alarms raised, not by a statistic.

One language

Every vital sign, lab and drug is translated to one name, one unit and one clock before any model sees it.

Only the past

Each value is used from the hour it became available. No information from the future, ever.

Judged by utility

A score that pays for every hour of early warning and charges for every false alarm and every missed patient.

Clinician in the loop

Clinicians set the alarm budget and get a reason per organ, so the alert is one they will act on.

What is different: existing alerts are one model, one hospital, one statistic. SIREN is a method that any hospital can plug its own records into and get a warning it can trust and tune.

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How SIREN works

1
Hospital records
vital signs, labs, drugs, nursing charts, as they are stored

2
One language
same name, unit and clock in every ward and hospital

3
One definition
Sepsis-3: suspected infection plus new organ dysfunction, applied identically

4
Hour by hour
at every hour the model sees only what was known by then

5
Risk + reason
risk of sepsis in the next 6 hours, and which organ is driving it

6
Alarm policy
the ward's alarm budget decides when a rising risk becomes a bell

What the clinician sees

A risk trajectory
the patient's six-hour sepsis risk over the stay, not a single number

An organ panel
kidneys, lungs, circulation, liver, clotting, brain: which one is moving

An alarm that respects a budget
two hours of rising risk, no repeat for six hours, only when something new was charted

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Prototype plan, feasibility and risks

What we build in the two-week sprint

Data	The open PhysioNet 2019 Sepsis Challenge set: 40,336 ICU patients from two US hospitals, hourly vitals and labs, free for research. No patient data of our own is needed.
Model	A simple hour-by-hour model (logistic regression or gradient-boosted trees) in Google Colab, trained with the strict rule that each hour sees only earlier hours.
Score	The challenge's published utility score: reward for warnings up to 12 h early, a cost for each false alarm, a penalty for each missed patient.
Screen	A Figma dashboard: risk trajectory, organ panel, alarm-budget slider. Tested with five clinicians using Google Forms feedback.

Risks we already know about

Alert fatigue	a budget of false alarms per patient-day, set with clinicians, is part of the design
Information from the future	every input is time-stamped by when it became available; we test the model with that rule enforced
Does not transfer	the challenge set has two hospitals; we train on one and test on the other before we claim anything
Privacy and regulation	open de-identified data only; a deployed version would be software as a medical device under SFDA rules
Clinician trust	a reason per organ and a tunable budget, tested with clinicians before any pilot

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Sustainability

SIREN keeps working because it is built as a method the hospital owns, not a black box it rents.

Lives in the hospital system

Runs on the electronic record the hospital already has, through standard interfaces (HL7 FHIR). No new devices, no new data entry.

Learns the local language once

The translation of each hospital's records is built once with its own informatics team and then reused for every future model and audit.

Watched after launch

Alarm rate, missed cases and clinician response are monitored every month; the model is retrained when they drift, which is also how the quality team reports it.

Funded by what it saves

Earlier antibiotics shorten ICU stays. Hospital quality budgets and national digital-health programmes already fund early-warning tools.

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Business Model

Key Partners

- Hospital informatics and quality teams
- ICU and emergency clinicians
- Ministry of Health digital-health programme
- University research groups
- Open ICU databases (PhysioNet)

Key Activities

- Translate each hospital's records to one language
- Train and audit the hourly model
- Co-design the alarm policy with clinicians

Key Resources

- Open ICU data for development
- Hospital EHR access via FHIR
- Clinical advisors; cloud compute

Value Proposition

- A sepsis warning hours earlier that works in this hospital, not only the one it was built in
- Fewer false alarms through a clinician-set budget
- A reason per organ, so the alert is actionable
- Transparent, auditable, hospital-owned

Customer Relationships

- Co-design with ICU teams
- Monthly performance reports to the quality committee
- Clinician feedback loop in the dashboard

Channels

- Inside the hospital EHR and nursing dashboards
- Pilot through a university hospital
- Ministry of Health innovation programmes

Customer Segments

- Public and private hospitals with ICUs
- Hospital quality and patient-safety departments
- Regional health clusters
- Later: wards and emergency departments

Cost Structure

Data translation per hospital (one-off) · cloud compute and monitoring · clinical and regulatory review · team salaries

Revenue Streams

Annual licence per hospital · implementation service per site · research and innovation grants during development · savings from shorter ICU stays fund adoption

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Development vision and next steps

Now · Sprint

Oct 26 to Nov 10

- Prototype on open PhysioNet data in Colab
- Figma dashboard
- Feedback from five clinicians

Finals · Nov 27–28

first test

- Train on hospital A, test on hospital B of the open set
- Report utility, false alarms per day, warning time
- Show the dashboard live

Phase 2 · 6 months

one hospital, silent

- Translate one Saudi hospital's records with its informatics team
- Run silently: alerts logged, not shown
- Compare with what happened

Phase 3 · 12 months

clinician pilot

- Alarm budget set with the ICU team
- Pilot on one unit with before-and-after measures
- SFDA pathway and ethics approval

Phase 4 · beyond

scale

- Second hospital with a different record system
- Wards and emergency department
- Regional cluster licence

Success at each phase is defined before the phase starts: a utility score above zero on the unseen hospital, a false-alarm rate inside the budget the clinicians set, and a warning that arrives before the first antibiotic order.

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Thank you

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